

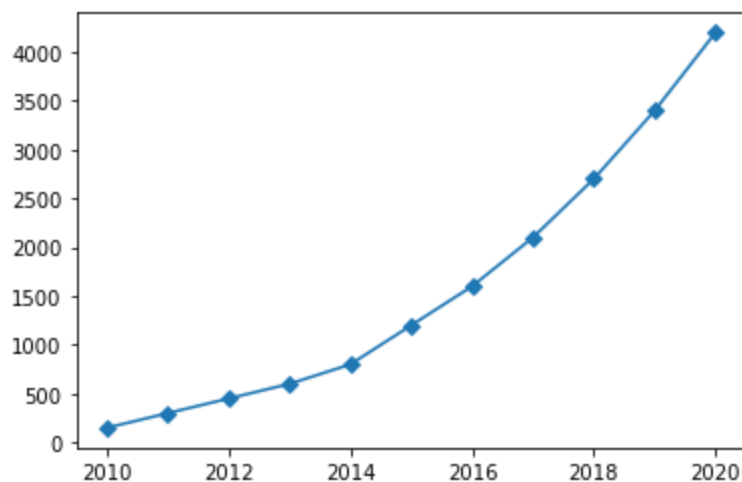
```
In [3]: import pandas as pd
import matplotlib.pyplot as plt
data={
    "Year":[2010,2011,2012,2013,2014,2015,2016,2017,2018,2019,2020],
    "Job Postings":[150,300,450,600,800,1200,1600,2100,2700,3400,4200]
}
df=pd.DataFrame(data)
df
```

Out[3]:

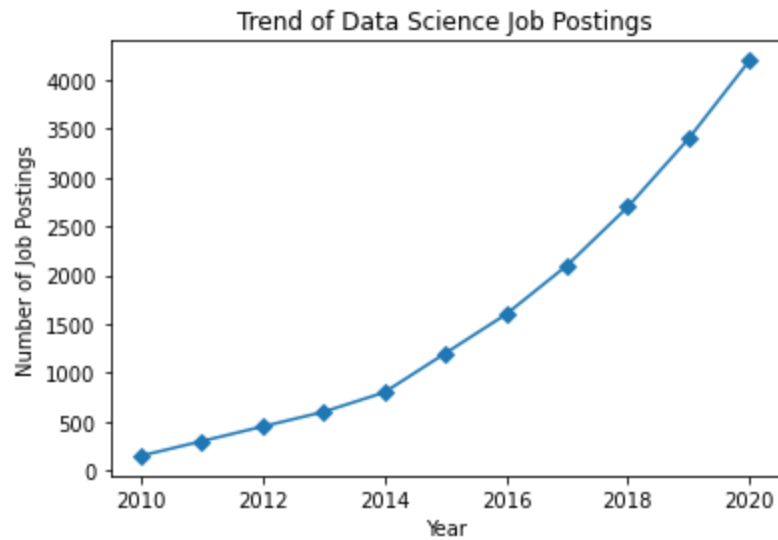
	Year	Job Postings
0	2010	150
1	2011	300
2	2012	450
3	2013	600
4	2014	800
5	2015	1200
6	2016	1600
7	2017	2100
8	2018	2700
9	2019	3400
10	2020	4200

```
In [10]: plt.plot(df['Year'], df['Job Postings'], marker='D')
```

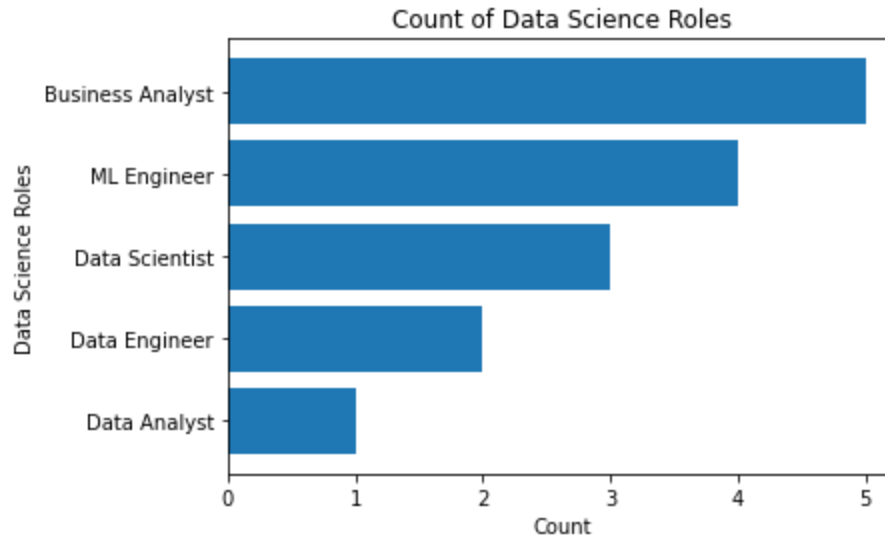
Out[10]: [



```
In [16]: import pandas as pd
import matplotlib.pyplot as plt
data={
    "Year":[2010,2011,2012,2013,2014,2015,2016,2017,2018,2019,2020],
    "Job Postings":[150,300,450,600,800,1200,1600,2100,2700,3400,4200]
}
df=pd.DataFrame(data)
plt.plot(df['Year'], df['Job Postings'], marker='D')
plt.title('Trend of Data Science Job Postings')
plt.xlabel('Year')
plt.ylabel('Number of Job Postings')
plt.show()
```



```
In [23]: import pandas as pd
import matplotlib.pyplot as plt
data={
    "Roles":["Data Analyst" , "Data Engineer" , "Data Scientist" , "ML Engineer"
    "Count": [1,2,3,4,5]
}
df=pd.DataFrame(data)
plt.barh(df["Roles"],df["Count"])
plt.title("Count of Data Science Roles")
plt.xlabel("Count")
plt.ylabel("Data Science Roles")
plt.show()
```



```
In [31]: import pandas as pd
data1={
    'Name': ['AAA', 'BBB', 'CCC'],
    'Age': [11,22,33]
}
data2="Hello World"
data3={"A":1,"B":2,"C":3}
df=pd.DataFrame(data1)
print("Unstructured Data: ",data2)
print("\nSemi-Structured Data: ",data3)
print("\nStructured Data:")
df
```

Unstructured Data: Hello World

Semi-Structured Data: {'A': 1, 'B': 2, 'C': 3}

Structured Data:

Out[31]:

	Name	Age
0	AAA	11
1	BBB	22
2	CCC	33

```
In [37]: from cryptography.fernet import Fernet
key = Fernet.generate_key()
f = Fernet(key)
token = f.encrypt(b"Rajalakshmi Engineering College")
token
f.decrypt(token)
key = Fernet.generate_key()
cipher_suite = Fernet(key)
plain_text = b"Rajalakshmi Engineering College."
cipher_text = cipher_suite.encrypt(plain_text)
decrypted_text = cipher_suite.decrypt(cipher_text)
print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
print("Decrypted Data:", decrypted_text)
```

Original Data: b'Rajalakshmi Engineering College.'

Encrypted Data: b'gAAAAABorS9IE1jRL-TwXdbap0Y2yDpTTThDXacOsaOH2mShtG6awcTYNCYmvRpMBrbdGs3tZTKuN7vs_MGvwZWS_ycNTd7-64MbGjbq3Szn1BvpmmJae1xOCd_VwL5_MAD3pxLZjp03q'

Decrypted Data: b'Rajalakshmi Engineering College.'